

PAMPANA SIVA SURYA RAM GOPAL

Software Engineer

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Location: Bengaluru, Karnataka, India

[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Python Full-Stack Developer skilled in Django, React, and Azure OpenAI with a focus on building scalable, AI-driven web applications. Experienced in backend APIs, frontend design, and database management. Completed automation testing training to enhance software quality and performance awareness. CCNA-certified with strong cloud and networking knowledge.

TECHNICAL SKILLS

Programming Languages:

Java, Python, JavaScript, HTML5, CSS3

Testing & Automation:

Selenium, Pytest, TestNG, JUnit, Automated Testing

Web Technologies:

React, Django

Cloud & AI:

AWS, Azure OpenAI, GPT-4, Azure AI Search

Databases:

MySQL, Azure Cosmos DB, SQL

Data Science:

Generative AI, Machine Learning, Data Analysis, LangChain, LangGraph

Tools & Platforms:

Git, VSCode, PyCharm, GitHub Copilot, Postman, REST APIs

WORK EXPERIENCE

Software Test Engineer Trainee

November 2024 - Present

Cisco Systems

Bengaluru, India

- Automated testing using Python with Selenium and Pytest to ensure software quality
- Developing a full-stack AI chatbot using Azure OpenAI, Django, MySQL, and Azure AI Search for real-time natural language query handling
- Integrating Azure AI Search for intelligent schema retrieval and dynamic SQL generation
- CCNA Certified: Proficient in networking, routing, switching, and IP fundamentals

KEY PROJECTS

AI-Powered Chatbot for Infrastructure Management (Cisco, 2024 – Currently Implementing)

Currently Implementing

- Multi-agent design with Azure OpenAI: GPT-4.0 primary agent and GPT-3.5-Turbo SQL agent.
- Intelligent query routing across RAG, SQL, and web-search using LangChain and LangGraph.
- RAG via Azure AI Search: document chunking, Azure embeddings, vector index for semantic retrieval.
- Autonomous SQL agent generates, validates, and executes queries on Azure MySQL Server and Azure Cosmos DB.

- Secure Azure environment using Azure OpenAI, Azure Blob Storage, Azure VNet, and private networking.
- Real-time, accurate, grounded responses; production-ready architecture.

Network-Based Pattern Searching System

2024

- Developed a **Network-Based Pattern Searching** project with **client-server architecture** to search for words in a file.
- Implemented **client-server communication** using **JSON requests** to query and return matching lines with line numbers.
- Designed a **Search class** to read, clean, and search file content efficiently, handling errors like "File Not Found."

Self-Fuel Dispensing Automated Framework Using RFID Cards

2023

- Developed and implemented an RFID-based self-fuel dispensing system, reducing operational errors and unauthorized access by 25%.
- Optimized transaction efficiency by 40% through advanced algorithms and real-time data processing.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering
Pragati Engineering College
Kakinada, Andhra Pradesh, India

August 2019 - June 2023

- CGPA: 7.87

CERTIFICATIONS

- **Data Science Certification** - Cisco Systems | 2025 - Generative AI Green Belt Program
- **CCNA (Cisco Certified Network Associate)** - Cisco Systems | 2024
- [**Cisco Certified DevNet Associate**](#) - Cisco Systems | 2023
- **Python Programming** - Completed comprehensive training program
- [**Automation Testing**](#) - Test Automation Training | 2024

ADDITIONAL INFORMATION

- **Interests:** AI/ML Technologies, Test Automation, Full-Stack Development, Cloud Computing
- **Professional Development:** Actively learning advanced Azure services and modern testing frameworks